

Laboratory 1

0. Read carefully and try the commands written in the first 3 pages of the tutorial.
1. Evaluate the number using floating point arithmetic: $\frac{1}{2}$; $e = \exp(1)$; $\sqrt{3}$ (`sqrt(3.)`); π (`evalf(Pi)`); $\sin(0.1)$.
2. Assign the following expression to a variable and then expand it:

a) $(x^2 + 2x - 1)(x^2 - 2)$

b*) $(x + n)^5$.

Unassign the used variables ($a := 'a'$ in Maple).

3. Factorize: $x^8 - 1$
4. Add the following rational expressions by applying the `factor` command:

$$\frac{2 \cdot x^2}{x^3 - 1} + \frac{3 \cdot x}{x^2 - 1}$$

5. Simplify the trigonometric expression (using `simplify(expression, trig)`):

$$\sin(x)^2 + \cos(x)^2$$

6. Evaluate using both `subs` and `eval` the expression $e^x + \ln(x)$ in $x = 1$.

7. Solve:

a) the equation $x^2 - 4 \cdot x + 3 = 0$ where x is the unknown;

b) the equation $x^2 \cdot y + 2 \cdot y - x = 0$ where x is a parameter and y is the unknown;

c*) the equation $x^2 \cdot y + 2 \cdot y - x = 0$ where y is a parameter and x is the unknown;

d) the equation $x - \cos(x) = 0$ where x is the unknown; (for this equation one can find only approximate roots. So, use `fsolve` in Maple, and `find_root` in Sage. Hint: it has 1 real root, and it belongs to the interval $[-1, 1]$)

e*) the equation $x^5 - 3 \cdot x^3 - 1 = 0$ where x is the unknown; (for this equation one can find only approximate roots. So, use `fsolve` in Maple. Hint: it has 3 real roots, all in the interval $[-2, 2]$)

f) the system of two equations $\{4x + 3y = 10, 3x - y = 1\}$ where x and y are the unknowns.

8. Assign to a variable f the function (not the expression) $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = e^x - \sin(x)$. Evaluate $f(0)$, $f(-1)$, $f(1)$. Calculate the first and second order derivatives of f (using both `D` and `diff`) and a primitive of f . Evaluate $\int_{-1}^1 f(x) dx$. Unassign f . Do not forget that for $f'(0)$ (the derivative of a function in a given point) you just have to write `D(f)(0)`.
9. Assign to a variable g the expression $e^x - \sin(x)$. Evaluate this expression in $x = 0$. Compute its first order derivative and then evaluate it in $x = 0$. Find a primitive. Evaluate the second order derivative of g and evaluate it in $x = 0$. Evaluate $\int_{-1}^1 g dx$.

10. Find $\lim_{x \rightarrow 0} \frac{\sin(x)}{x}$ and $\lim_{x \rightarrow \pi} \frac{\cos(x)+1}{x-\pi}$.
11. First plot the graph of $f(x) = \sin(x)$ by hand on paper. Then plot it using Maple in each of the intervals: $[0, 2\pi]$, $[-4\pi, 4\pi]$, $[-100, 100]$, $(-8, 8)$.
12. First plot the graph of $f(x) = \frac{1}{x}$ by hand on paper. Then plot it using Maple in the interval $[-1, 1]$.
13. Plot the planar curve of parametric equations $x = 2 - t^2$, $y = t - t^3$, $t \in [-2, 2]$.
14. First plot by hand on paper the planar curve of parametric equations $x = \cos(t)$, $y = \sin(t)$ in each of the intervals: $[0, \frac{\pi}{3}]$, $[0, \pi]$, $[0, \frac{3\pi}{2}]$, $[0, 2\pi]$, $[0, 4 \cdot \pi]$. Then do the same using Maple. What do you notice?
- 15*. Plot the planar curves of parametric equations
 - a) $x = 2 \cdot \cos(\frac{t}{3})$, $y = 2 \cdot \sin(\frac{t}{3})$
 - b) $x = \cos(4 \cdot t)$, $y = \sin(4 \cdot t)$
 in different intervals at your choice. What do you notice?
16. Write by hand on paper the implicit equations of a circle centered in $(9, 10)$, an ellipse and a parabola. Then plot using Maple at least a circle, an ellipse and a parabola.
17. Plot the planar curves of implicit equations:
 - a) $x^2 - 2 \cdot x \cdot y - y^2 = 1$
 - b*) $y^3 + y^2 - 5 \cdot y - x^2 = -4$.
 You have to choose properly a rectangle where to see the curve.
18. First draw by hand on paper the graph of the function $H(x, y) = x^2 + y^2$. Then plot it in 3d using Maple. For the variable (x, y) choose a rectangle centered at $(0, 0)$. What remarkable planar curves are the level curves of H ?